The Current Strength of the AI revolution

A dive into the moving parts of the AI supply Chain.

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In the quest to simplify the world to a comfortable level, the byte-sized pieces of information are shaping the opinions of the masses. Most want to snack on pre-cooked, easy-digestible and highly processed pieces of “truths” rather than spen time in the kitchen assembling nutritious raw facts to a delicious and complex me insights.

I am not any different. I also prefer the snacks over the hard work of analysis, but the same time, I am driven by an urge to understand more. I want to understand mechanics behind what is happening.

While it is a lot easier to adopt an opinion served without any research, I keep my mind in aeroplane mode until the data lands. It is not easy, but it is necessary to generate insights.

Most of my research is focused on and around the Semiconductor supply chain. to truly generate insights, I need to venture outside my comfort zone and into th broader realm of other knowledge areas where I am less competent.

The risk for me is minimal, but for my ego, it is vast. If I make a mistake, the inter will keep it forever. I really should stay in my little, comfortable garden and prune ego, never to be higher than the hedges.

Well, that was fun to write, and it's too late. I am too far into the weeds to stop n

The business topic above all others is Artificial Intelligence, and as I am curious a the business of semiconductors, the orange fell into my lap. While I am no exper

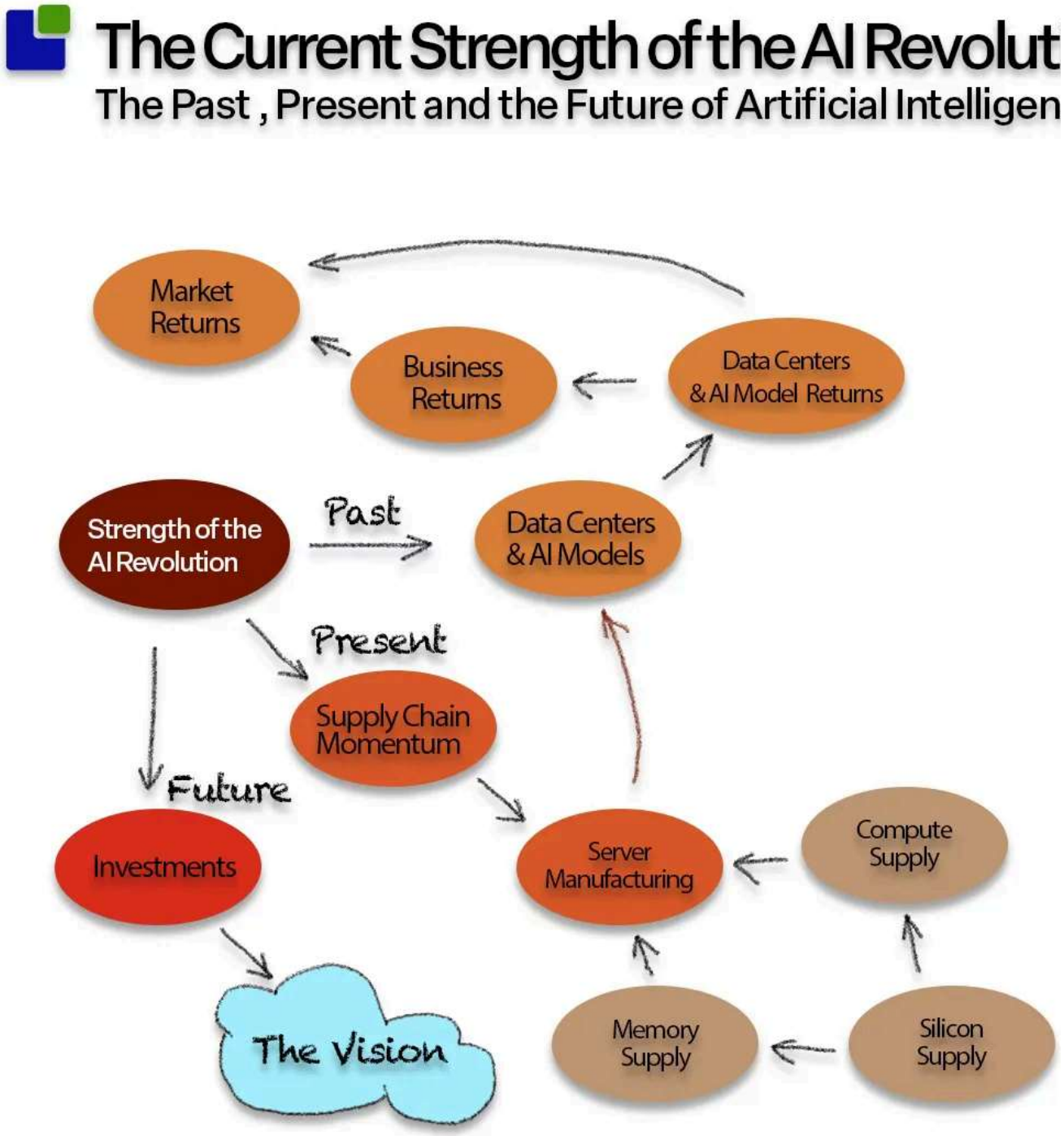
I have some of the puzzle pieces and have an urge to use them to uncover insigh

Since the first Nvidia AI quarter just over two years ago, there have been questio about how long this will last. Is it a bubble, and when will it burst?

While not new, concerns have intensified over the last month following the underwhelming presentation of ChatGPT 5 in a dorm-style setting.

The most recent result from Nvidia was sufficiently good (we have grown accustc to stellar results) to avert disaster, and the balloon remains airborne.

Before answering, it is essential to understand what the question is, and that brin back to the concept of complexity. My loosely formed question, “What is the stre of the AI revolution?” has many complex elements. Fortunately, I decided many r ago that complexity is my friend.



To get a grip on the question, begin to scribble in my notebook. For this questior use the concept of time. While the stock market returns have an element of "futu (for stock analysts, this means next quarter), the tangible results are based on the as is the case with business returns.

I will work my way through the different subquestions, focusing on business data usual, while leaving the technical analysis of the AI models to more capable indiv

I will not be able to cover all the different topics in one post, but I will follow up v subsequent post that continues the research.

Market Returns

The concept of a bubble is primarily associated with stock market valuations, alth variations of this concept exist. I know this is of utmost importance to many, as tl lifestyle and retirement depend on it. It is not different for me. I own stocks in mc the companies I write about, but I don't trade very often, and I never waste my ti talking stocks up or down.

The stock market return and the business returns are two different topics. While s market returns should depend on business returns, they sometimes become disconnected, as may be the case right now.

AI Leaders Financials

Company	EV ($ Bn)	Market Cap ($ Bn)	TTM Sales ($ Bn)	Price (8/14)	Price as % of 52-Week High	EV/TTM EBITDA	EV/TTM SALES
Nvidia	$4,419.2	$4,462.6	$246.4	$182.9	99.9%	49.9x	17.9x
Microsoft	$3,926.0	$3,908.4	$297.7	$525.8	98.2%	23.3x	13.2x
Apple	$3,415.4	$3,446.7	$472.1	$232.3	89.7%	24.1x	7.2x
Amazon	$2,510.0	$2,451.3	$701.5	$229.8	95.0%	17.0x	3.6x
Alphabet	$2,401.6	$2,458.1	$429.5	$202.7	98.2%	16.8x	5.6x
Meta	$1,979.8	$1,977.3	$190.4	$787.1	99.6%	20.5x	10.4x
Tesla	$1,069.2	$1,092.0	$99.0	$338.6	70.6%	94.2x	10.8x
Tencent	$706.3	$689.5	$0.7	$75.3	100.0%	18.0x	6.2x
SAP	$343.1	$343.5	$41.0	$279.6	90.0%	28.5x	8.4x
IBM	$274.2	$221.8	$53.7	$238.1	80.8%	19.4x	5.1x
AMD	$294.4	$296.3	$37.7	$182.6	86.4%	55.8x	7.8x
Salesforce	$219.9	$225.3	$30.3	$235.7	64.1%	19.0x	7.3x
Alibaba	$233.8	$296.7	$127.1	$15.5	84.3%	-	1.8x
Qualcomm	$175.6	$170.8	$43.0	$158.3	69.7%	12.8x	4.1x
Adobe	$149.9	$149.1	$19.0	$351.5	55.4%	16.5x	7.9x
Intel	$129.2	$97.2	$52.2	$22.2	43.8%	-	2.5x
CoreWeave	$65.7	$52.3	$2.6	$107.0	58.3%	-	25.0x
Baidu	$8.0	$31.3	$17.4	$89.2	60.9%	2.0x	0.5x

Source: AlphaSense; Bloomberg; Capital IQ

From: Artificial Intelligence H1 2025 Global Report

EV ($ Bn): Enterprise Value = market cap + debt – cash. A measure of what it v cost to buy the entire company.

Market Cap ($ Bn): Current stock price × number of shares outstanding.

***TTM Sales ($ Bn)**: Trailing Twelve Months Sales — total revenue over the last*

***Price (8/14)**: Share price on August 14, 2025.*

***Price as a Percentage of 52-Week High: Indicates** how close the stock is to its peak.*

***EV/TTM EBITDA**: Valuation multiple = EV ÷ EBITDA (earnings before interest, depreciation, amortisation). High multiples indicate that a company is expen relative to its profits.*

***EV/TTM Sales**: Valuation multiple = EV ÷ Sales. Used primarily when profits ar or negative.*

There are many ways to value companies, but ultimately, valuations must be tied the ability to deliver future free cash flow, which in turn is tied to earnings. The E EBITDA for global stocks is between 10 and 15, implying that a lot of the AI stock if not in bubble territory, then certainly frothy.

Most concerns have been related to business returns, but they have not yet had material impact on AI stock valuations.

Business Returns

Apart from the failed launch of ChatGPT 5 and consumer preference for ChatGPT there are numerous negative stories about the business returns of AI projects. I h collected a couple, but there are many more.

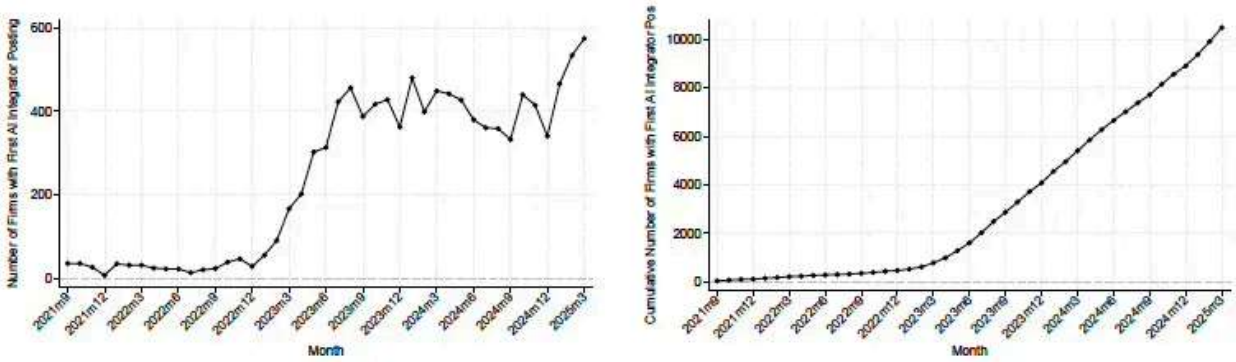
- MIT's NANDA study reveals only 5% of AI pilot programs succeed Link
- An analysis notes a widening gap between AI innovation promises and actua business returns—only 13% of enterprises report meaningful impact, with infrastructure and regulation costs hindering progress. Link

- Goldman Sachs and Deloitte estimate that up to 30% of AI projects stall due poor data, unclear value, or rising costs; RAND echoes an 80% AI failure rate, casting doubt on trillion-dollar capex plans. Link

This is not unusual for new technology, nor is it uncommon that corporate chang programs fail in general. The often-quoted 70% failure rate is most certainly too but I will use the almost accurate term of: A lot!

Behind the scenes, the story is somewhat different. The models that businesses a consumers get access to are far from the best offered by the AI companies, as th expensive to use, even with a declining token price. While the ChatGPT 5 launch failure, the model itself scored highly on the Mensa test and generally receives g feedback from AI experts.

New research utilises LinkedIn data to shed light on the corporate use of AI throu job titles.



(a) Flow (b) Cumulative

Figure 1: Timing of Firms' First AI-Integrator Vacancies

Notes: Panel (a) shows the monthly number of firms posting their first AI-integrator vacancy, while Panel (b) reports the cumulative total, covering the period from September 2021 to March 2025. See Section 3.2 for details on the identification of AI-integrator postings.

The number of first AI integrator vacancies has been steady since mid-2023, indi that AI implementation is now an ongoing process rather than a step function, a more than 10,000 companies now have AI integration underway, according to th research. There is much AI revenue to come.

The same research also reveals an interesting fact about the changes in job posti seniority.

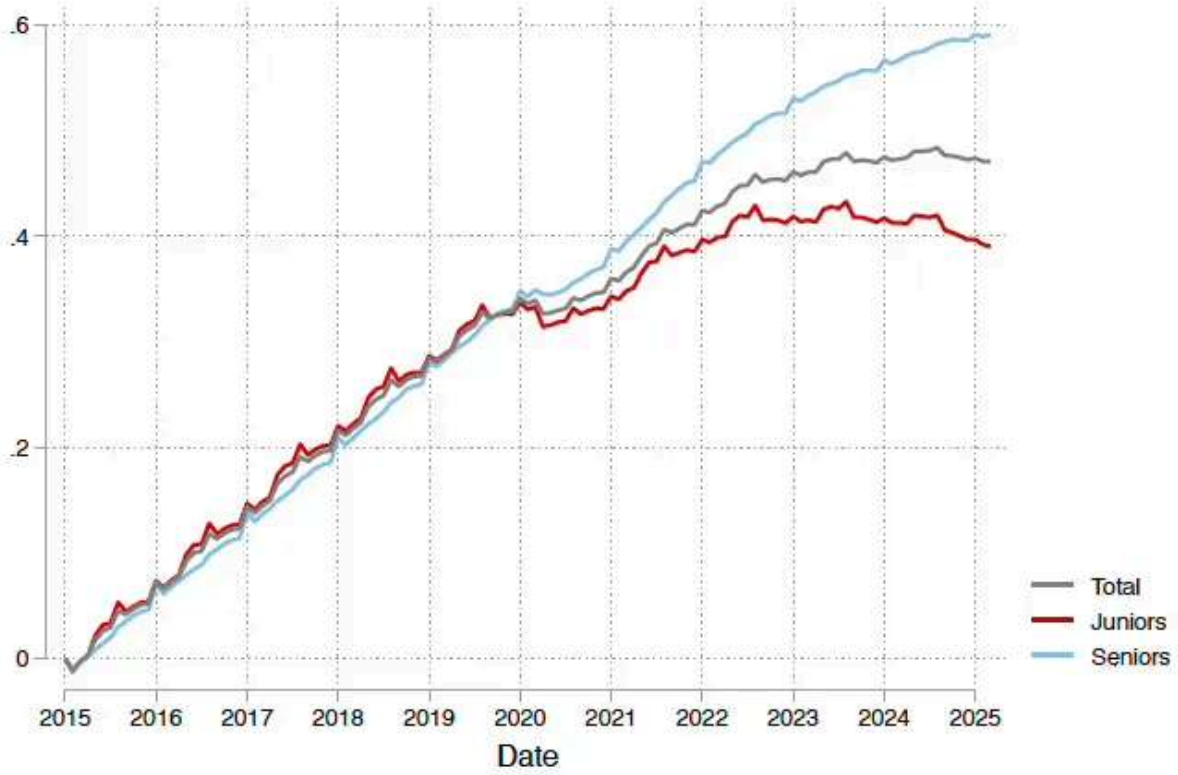


Figure 2: Time Series of Junior and Senior Employment in Sample Firms

Notes: This figure plots the average number of junior-level workers and senior-level workers our sample of firms over time, normalized to 1 in January 2015. We define "junior" workers those in Entry- or Junior-level positions, and "senior" workers as Associate level and above (s Section 3.1 for details).

During and after the COVID-19 pandemic, there was a shift in the employment trajectories of Junior and senior employees, who had been on similar paths since

As can be seen, the paths diverge around mid-2022, coinciding with the introduc of LLMs, and the gap is now widening. This suggests that the AI is not only repla employees but mainly junior employees. The narrative of LLMs making everybod specialist has to be replaced with the story that LLMs need to report to a speciali have all seen the hallucinations that make this story more plausible.

While agentic AI could change that once again, productivity gains are being mad right now, according to this data. This is impacting employment on a larger scale

the Semiconductor industry, but in my sector, I have the data to prove it. Read m about the employment below:

Bleak Blue Badges

CLAUS AASHOLM · JULY 28, 2025

Read full story →

While our billionaire friends talk about us not having to work so hard anymore, tl are less informative about how we are going to make money. Likely, we will be tc that it is trickle-down economics as usual. If we are unhappy, we can still vote, ev though social media decides elections. I am digressing - back to the question.

The returns on using AI so far have been focused on business benefits, with a pri emphasis on cost savings in a corporate environment. While consumers experim with LLMs, major shifts are underway.

According to Epoch AI, a top-of-the-line Gaming GPU can now run a frontier mo from 6 to 12 months ago, meaning AI will soon be at the consumer edge and be change our physical environment. If you have a Nokian love of buttons and smal screens, you'd better brace yourself. All future consumer products will feature ne modalities: To see/hear and respond.

Additionally, the physical manifestation of AI in the form of general-purpose robo is still in its early phase of development.

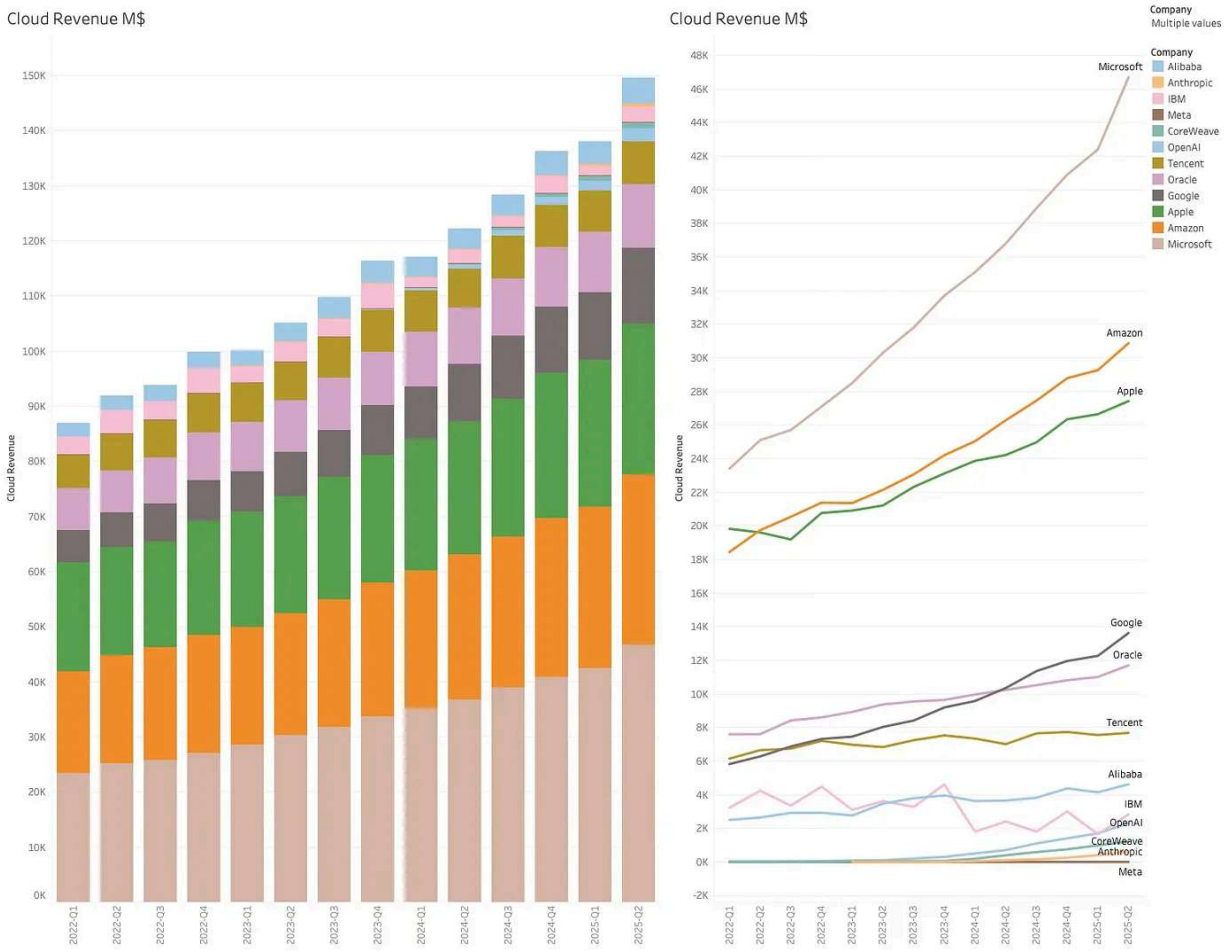
The market might crash, but the business returns will continue to drive the implementation forward. The generally accepted narrative is that employees will lose their jobs to AI, but lose their jobs to another employee using AI can be exte to companies.

Companies that utilise AI will outcompete those that do not. It will be challengin some leaders to understand that AI is not an add-on to the product that needs tc justified, but a change to the fabric of the product itself.

The returns of the Data Centres and AI model compani

The direct revenue from AI is not readily separable from cloud revenue, and I will attempt to do so, as it makes little sense. While traditional cloud computing still accounts for the majority of current data centre capacity, investments in AI have dwarfed those in CPU-based systems.

The cloud revenue is also in rapid transformation towards AI, which is exclusively driving the growth. It sometimes sounds like the large cloud companies are spen bucketloads of capital expenditure (CapEx) without a return. Still, the reality is tha cloud revenue is growing at a healthy rate.



While Meta is a key player in AI, the social media company has yet to monetise it capabilities directly, as Twitter (a prior usable version of X) has done with Grok. Microsoft is the clear leader in cloud revenue and continues to accelerate its grov

The native AI companies are still far from having evolved their full commercial potential, with OpenAI leading the pack. Sam Altman is pursuing what have beer

called “Landgrab” deals, giving an entire country access to the extended model. T date, discussions have been held with the UK, India, and several countries in the Middle East.

In the field of AI model development, there are indications of self-improving AI n and models for training. There is also a trend towards smaller, higher-specialised models that require smaller training sets.

After the launch of GPT5, the Agentic workload doubled, and reasoning increase times.

Rumours of Meta going into an AI hiring freeze after the frenzy are counterbalan by Microsoft launching a most-wanted list of AI researchers.

While it is impossible to gain a good understanding of how much compute is ne the key constraint to the data centre investments will be the revenue generated b While the current revenue is based on investments made over a year ago, a plate cloud and AI revenue could limit capital expenditures.

The revenue of the large cloud companies is growing at a CAGR of 8.6% in the p shown, while the cloud revenue is growing at 17%.

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